

SOMA: EFFICIENT MULTI-TURN LLM SERVING VIA SMALL LANGUAGE MODEL

Anonymous authors

Paper under double-blind review

ABSTRACT

Large Language Models (LLMs) are increasingly deployed in multi-turn dialogue settings where preserving conversational context across turns is essential. A standard serving practice concatenates the full dialogue history at every turn, which reliably maintains coherence but incurs substantial cost in latency, memory, and API expenditure, especially when queries are routed to large proprietary models. Existing approaches often struggle to balance the trade-off between response quality and efficiency. We propose a framework that exploits the early turns of a session to estimate a local response manifold and then adapt a smaller surrogate model to this local region for the remainder of the conversation. **Concretely, we train soft prompts that push the surrogate small model away from the large model’s responses, stabilized by anti-degeneration control, and score each early turn by how much agreement survives this push; localized LoRA fine-tuning then distills the robustly anchored turns, so the surrogate runs without prompts at inference. A gate hands the session to the surrogate and rolls back on drift.** We further provide a theoretical analysis for key components in SOMA. **Extensive experiments on six multi-turn benchmarks show that SOMA preserves the original model’s responses and reduces serving cost for medium-to-long, locally coherent sessions.** The source code is provided at: <https://anonymous.4open.science/r/SOMA-4175>.

1 INTRODUCTION

Large language models (LLMs) such as the GPT series (Brown et al., 2020; Achiam et al., 2023), LLaMA (Touvron et al., 2023), Claude (Anthropic, 2023), and DeepSeek (Guo et al., 2025) power machine-learning-as-a-service (MLaaS) applications from chat assistants to code generation (Park & Kulkarni, 2023; Dong et al., 2023; Liu et al., 2024; Li et al., 2026; Yu et al., 2026), and *multi-turn LLM serving*, which covers extended interactions between humans and LLMs or among LLM agents, has become a key research focus (Yi et al., 2024; Li et al., 2025b; Zhao et al., 2025). Serving such interactions efficiently remains hard: most serving systems are stateless and resend the entire conversation history with every new query (Ananda, 2025; Moon, 2025), so computation, latency, and cost all grow as the conversation lengthens.

Prior work follows two routes. *Single-model methods* compress the dialogue history (Wang et al., 2025; Chen et al., 2024b; Xiao et al., 2024), retrieve memory from external modules (Melz, 2023; Gutiérrez et al., 2024), or reuse attention computation (Gao et al., 2024; Jeong & Ahn, 2025; Anthropic, 2024); they still run the large model at every turn, so cost, latency, and GPU usage stay high, and they often truncate the context that long dialogues depend on. *Multi-model methods* route easy queries to small models and escalate hard ones (Behera et al., 2025; Schick et al., 2023; Ding et al., 2024; Shnitzer et al., 2023), but small models generalize poorly across dialogue complexity, and switching adds overhead. Moreover, LLMs rely heavily on early turns (Xiao et al., 2024; Laban et al., 2025), which makes coherence harder to maintain as a dialogue grows.

Is it possible to achieve an efficient, context-aware multi-turn LLM serving framework that avoids recomputing the full history at every turn while preserving response quality?

To answer this question, we examine real-world multi-turn dialogues across domains and find a long-tail distribution of token counts over turns: early turns are substantially longer, while later turns shrink. This matches the intuition that early turns carry **the substantive openings that set the issues and anchors of a conversation**, whereas later turns are often brief follow-ups (He et al., 2018; Stolcke

et al., 2000). The trend suggests that a smaller model might serve the short later turns cheaply. The bottleneck, however, is the “big head”: most grounding is established early, and later turns depend on it, so a small model that behaves reasonably at first drifts from the large model as the dialogue progresses. The small model must therefore approximate the larger model’s behavior within the local region of its output space that the prior dialogue induces, which we formulate as a local manifold approximation problem (Section 2).

Building on these insights, we present SOMA (Soft-prompts for lOcal Manifold Approximation), which adapts a small language model to the local behavior of a large one, conditioned on the early turns of a session, through a three-stage pipeline: (1) *soft-prompt tuning*, which pushes the small model away from the large one within the local region and scores each early turn by how much agreement survives; (2) *localized fine-tuning*, which aligns the small model with the large one on the early turns, weighted by these scores; and (3) *efficient inference*, which serves later turns from a compressed context and rolls back to the large model when the topic drifts. Our contributions are:

- **Long-tail pattern in multi-turn dialogues.** The first few turns of a dialogue concentrate heavy context, while later turns are shorter yet more dependent on earlier ones, so a small, cheap model could serve the later turns if it is given the accumulated context.
- **SOMA: Efficient multi-turn serving.** Adversarial soft prompts score how robustly the surrogate reproduces the original model on each early turn, a localized LoRA adapter distills the robustly anchored behavior, and a cosine gate handles switching and rollback at prompt-free inference time.
- **Theory analysis and empirical evaluation.** We give a held-out acceptance bound for switching, a per-session cost model, and a coverage bound for random prompt candidates, each with its assumptions stated, and we validate SOMA on six multi-turn benchmarks and two model families.

2 PRELIMINARIES

2.1 NOTATIONS

A multi-turn dialogue prefix of length k is $\mathcal{D}_k = \{(q_1, a_1), \dots, (q_k, a_k)\}$, where q_t is the user query at turn t and a_t is the model response. We write F for the original model, a black-box LLM as in MLaaS, and G for the small surrogate; the response of $M \in \{F, G\}$ at turn t is a_t^M . We write $S_t = \mathcal{D}_{t-1} \oplus q_t$ for the dialogue state at turn t , i.e., the history followed by the new query. A soft prompt of length L is a matrix $\mathbf{P} \in \mathbb{R}^{L \times d}$. Appendix A lists all symbols used in the paper.

2.2 EXPLORING TOKEN-TURN PATTERNS IN MULTI-TURN LLM DIALOGUES

Standard serving processes each new request together with the entire history, so the cost of a turn grows with the length of the conversation. The amount of *new* information in a turn, however, need not grow at all. We therefore first study token usage across turns.

We examine four representative settings, which together cover different speaker roles, task goals, and interaction structures: *ShareGPT* (Chen et al., 2024a) for open-ended human-LLM chat, *ReMeDi* (Yan et al., 2022) for doctor-patient consultation, *Craigslist Bargain* (He et al., 2018) for negotiation, and *Multi-Character* (agent-lans, 2024) for multi-party role-play. For each dataset, we compute the average token count at turn t and normalize it by the average of Turn 1.

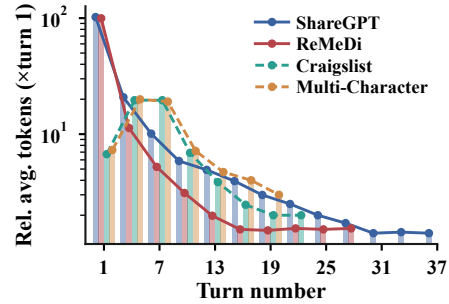


Figure 1: Average token count per turn, normalized by Turn 1, on four dialogue datasets: usage drops after the early turns and forms a long tail.

Figure 1 shows a consistent long-tail pattern. Early turns are token-heavy because they introduce the task, the topic, the constraints, and the speaker’s intent; later turns are much shorter and settle at a low, stable level. ShareGPT and ReMeDi peak at the opening turn, whereas Craigslist and Multi-Character peak shortly after it, once the negotiation or role-play is set up. Shorter does not mean independent: many later turns are local updates to an established dialogue state, so the marginal new input shrinks while standard serving keeps re-processing the full prefix. This mismatch motivates an adaptive

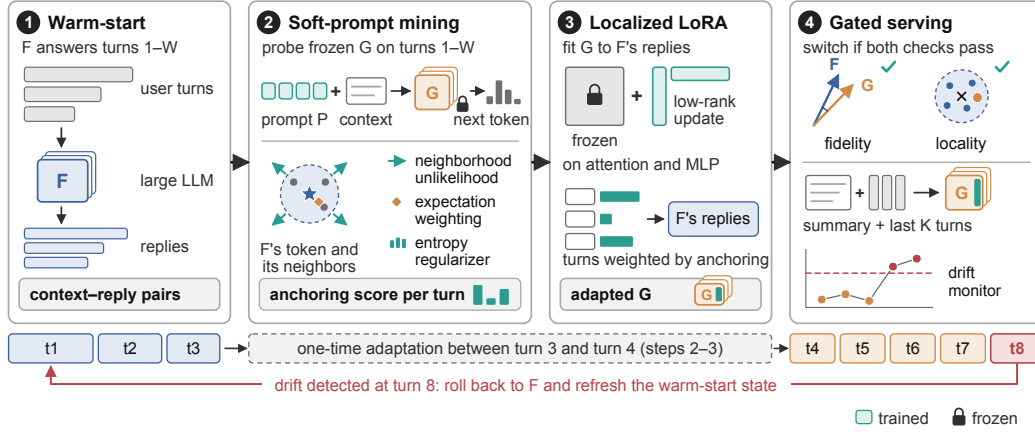


Figure 2: Overview of SOMA. (1) F serves the warm-start turns; (2) an adversarial soft prompt on the frozen surrogate G scores how robustly G still reproduces F on each turn; (3) a LoRA weighted by this score adapts G to F 's replies; (4) G takes over after a fidelity and locality check and rolls back to F on drift. Bottom: the matching session timeline, aligned with the four steps.

strategy: let the original model establish the early dialogue state, and hand subsequent turns to a cheaper surrogate for as long as they stay within that state.

2.3 LOCAL MANIFOLD APPROXIMATION FOR MULTI-TURN DIALOGUES

The long-tail pattern creates an opportunity for surrogate serving, but a naive substitution is unreliable. A later turn is short precisely because much of its meaning is supplied by the preceding dialogue, so a surrogate that is not aligned with the original model under that prefix produces fluent responses that quietly drift from the intended context.

We make this challenge precise through a local view of the session. Given a prefix $\mathcal{D}_k$, let Q_k be the distribution of the dialogue states $S_t = \mathcal{D}_{t-1} \oplus q_t$, $t > k$, that the session visits afterwards; we call its support the local region, or local manifold, induced by the prefix. The hidden spaces of F and G differ in dimension and have no canonical correspondence, so we compare the two models through their responses: a fixed sentence encoder ϕ embeds both, and $\text{Gap}_G(S) = 1 - \cos(\phi(a^G(S)), \phi(a^F(S))) \in [0, 1]$. The goal is a local rather than a global match, which is what makes a much smaller surrogate viable.

Problem 2.1 (Local Manifold Approximation for Multi-Turn Interactions). *Given a prefix $\mathcal{D}_k$ and the original model's responses $a_1^F, \dots, a_k^F$, learn a surrogate $G \in \mathcal{G}$ that minimizes the expected discrepancy on the turns that follow:*

$$\min_{G \in \mathcal{G}} R_k(G) = \mathbb{E}_{S \sim Q_k} [\text{Gap}_G(S)].$$

Only the first k turns are observed, so the problem is solvable only when they supervise the later ones. The local view makes this testable: while later states stay close to the warm-start states, the warm-start gap predicts the later gap, so post-switch similarity should rise with the warm-start length and saturate, and it should drop when the topic shifts (Figures 3(c) and 6). In our data, goal-anchored dialogues such as medical consultations and negotiations meet this premise better than reasoning-heavy or multi-party dialogues (Section 5).

3 SOMA: LOCAL MANIFOLD APPROXIMATION VIA SOFT PROMPTS

Figure 2 summarizes SOMA in four steps. (1) The original model F serves the first W turns and thereby establishes the dialogue state. (2) Adversarial soft-prompt mining measures, for each warm-start turn, how much of the agreement between the surrogate G and F survives a perturbation designed to remove it (Sections 3.1–3.2). (3) A lightweight LoRA adapter distills F 's responses,

weighted by this score, and (4) a semantic gate hands the session to G , with rollback to F when the dialogue leaves the learned region (Section 3.3).

3.1 INITIALIZATION

We initialize a soft prompt $\mathbf{P} \in \mathbb{R}^{L_p \times d}$ on the surrogate G by sampling each row i.i.d. from $p_\ell \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, where $\sigma > 0$ sets the initialization scale, and let $\mathbf{E} = [\mathbf{e}_v]_{v \in \mathcal{V}} \in \mathbb{R}^{d \times |\mathcal{V}|}$ be the surrogate embedding matrix, which also defines the semantic neighborhoods used below.

With $\mathcal{D}_{t-1}$ the text history before turn t and q_t the user query, the mining target is the response that F already produced while serving the warm-start turn, and the surrogate is probed with the soft prompt prepended at its embedding layer:

$$a_t^F = F(\mathcal{D}_{t-1} \oplus q_t), \quad a_t^G = G(\mathbf{P} \oplus_{\text{emb}} \mathbf{E}(\text{tok}_G(\mathcal{D}_{t-1} \oplus q_t))).$$

Here $\text{tok}_G(\cdot)$ is the tokenizer of G , $\mathbf{E}(\cdot)$ maps tokens to embeddings, and $\oplus_{\text{emb}}$ concatenates along the sequence. Both models see the same dialogue state, and only the surrogate is perturbed: F is called once per warm-start turn, never with the prompt, its responses stay fixed during optimization, and gradients flow only through $\mathbf{P}$ on G (Appendix G).

3.2 ADVERSARIAL SOFT-PROMPT MINING

We learn $\mathbf{P}$ by minimizing a differentiable objective that pushes G away from F on the local dialogue context. The objective has three parts: a token-level semantic divergence loss, an expectation-weighted distribution-level term, and an anti-degeneration regularizer.

Semantic divergence loss. An unlikelihood-style loss (Welleck et al., 2019) discourages G from assigning high probability to the tokens produced by F . An exact-token penalty alone is too narrow, since G could avoid the teacher token yet still favor close paraphrases, so we penalize a semantic neighborhood in the surrogate embedding space.

Definition 3.1 (Semantic Neighborhood). For token $u \in \mathcal{V}$, let $\mathcal{N}_k(u)$ be the set of the k tokens in $\mathcal{V} \setminus \{u\}$ whose embeddings have the largest cosine similarity to e_u .

At turn t the original model produces a_t^F , which we tokenize with the surrogate tokenizer, $y_t^F = \text{tok}_G(a_t^F) = (y_{t,1}^F, \dots, y_{t,T_F(t)}^F)$, so that teacher output and surrogate distribution share one vocabulary. Each teacher token spreads temperature-weighted mass over itself and its k nearest neighbors in the surrogate embedding space:

$$s_\tau(v | y_{t,i}^F) = \frac{\exp(\cos(e_v, e_{y_{t,i}^F})/\tau)}{\sum_{u \in \{y_{t,i}^F\} \cup \mathcal{N}_k(y_{t,i}^F)} \exp(\cos(e_u, e_{y_{t,i}^F})/\tau)}, \quad \tau > 0.$$

Let $S_t = \mathcal{D}_{t-1} \oplus q_t$. With causal masking, a single forward pass of G on the prefix S_t and the teacher answer prefix gives the next-token distributions $\{\Pi_{t,i}(\mathbf{P})\}_{i=1}^{T_F(t)}$, where $\Pi_{t,i}(\mathbf{P}) = \pi_G(\cdot | S_t; \mathbf{P}, a_{t,<i}^F)$. The semantic divergence loss is

$$\mathcal{L}_{\text{sem}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t) = \frac{1}{T_F(t)} \sum_{i=1}^{T_F(t)} \sum_{v \in \{y_{t,i}^F\} \cup \mathcal{N}_k(y_{t,i}^F)} s_\tau(v | y_{t,i}^F) \left[-\log(1 - \Pi_{t,i}(\mathbf{P})[v]) \right]. \quad (1)$$

Minimizing Eq. 1 removes the agreement of G with F on exact tokens and on nearby semantic alternatives, so agreement expressed through a paraphrase cannot survive the probe.

Expectation-weighted semantic divergence. The neighborhood loss can still miss distribution-level alignment: G may lower the probability of a teacher token and its neighbors while spreading mass over many other tokens with similar meaning. We therefore compare the teacher token’s embedding with the expected embedding of the surrogate’s next-token distribution, $\bar{e}_{t,i} = \mathbf{E}\Pi_{t,i}(\mathbf{P}) = \sum_{v \in \mathcal{V}} \Pi_{t,i}(\mathbf{P})[v]e_v$. We therefore upweight the neighborhood penalty whenever the expected embedding still points toward the teacher token, using the weight $w_{t,i}(\mathbf{P}) =$

$1 + \lambda_{\text{exp}} \text{clip}\left(\cos(\bar{e}_{t,i}, e_{y_{t,i}^F}), 0, 1\right)$ with $\lambda_{\text{exp}} \geq 0$. The final semantic mining loss is

$$\mathcal{L}_{\text{sem_exp}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t) = \frac{1}{T_F(t)} \sum_{i=1}^{T_F(t)} w_{t,i}(\mathbf{P}) \sum_{v \in \{y_{t,i}^F\} \cup \mathcal{N}_k(y_{t,i}^F)} s_\tau(v | y_{t,i}^F) \left[-\log(1 - \Pi_{t,i}(\mathbf{P})[v]) \right]. \quad (2)$$

Thus the loss blocks local copying through the neighborhood term and detects broader semantic shadowing through the expectation weight.

Anti-degeneration regularizer. Mining can collapse if G piles probability onto a few high-frequency tokens, yielding bland continuations that **suppress F 's tokens trivially and make the per-turn scores uninformative** (Li et al., 2023; Holtzman et al., 2019; Meister et al., 2023). We therefore preserve entropy near the prompt-context boundary; reusing the logits of the same forward pass, define

$$H_{\text{tail}}(\mathbf{P}; t) = \frac{1}{K} \sum_{i \in \text{tail}(t, K)} \left[- \sum_{v \in \mathcal{V}} \Pi_{t,i}(\mathbf{P})[v] \log \Pi_{t,i}(\mathbf{P})[v] \right].$$

The complete prompt-mining objective for a minibatch of warm-start turns $\mathcal{B}$ is

$$\mathcal{J}(\mathbf{P}) = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} [\mathcal{L}_{\text{sem_exp}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t) - \beta H_{\text{tail}}(\mathbf{P}; t)] + \lambda_P \|\mathbf{P}\|_F^2. \quad (3)$$

We minimize Eq. 3 with AdamW and gradient clipping while all weights of G stay frozen. Neighbors come from an ANN index over normalized token embeddings, and $\bar{e}_{t,i}$ uses a top- m truncation of $\Pi_{t,i}(\mathbf{P})$, so the per-step overhead is linear in the number of teacher-prefix positions and scales with $k + m$ rather than with the vocabulary size $|\mathcal{V}|$.

3.3 LOCALIZED FINE-TUNING, SWITCHING, AND ROLLBACK

Each warm-start turn receives an anchoring score, its largest residual mining loss over the candidate prompts, $r_t = \max_{\mathbf{P} \in \mathcal{P}_{\text{cand}}} \mathcal{L}_{\text{sem_exp}}(\mathbf{P}; \mathcal{D}_{t-1}, q_t)$. Since $\mathcal{L}_{\text{sem_exp}}$ grows with the probability that G assigns to F 's tokens and their neighbors, a large r_t means that G still predicts F 's response after a probe has pushed it away, so the response is anchored in the dialogue state rather than in turn-specific details. Unlike raw agreement, r_t counts only the agreement that the probe cannot remove, which separates robust matches from incidental ones. The training set $\mathcal{H}$ holds the warm-start turns with F 's responses, weighted by r_t : anchored responses carry the topic, constraints, and style that later turns share, whereas fragile ones carry turn-specific content that can dominate a fit on W examples, in the spirit of small-loss selection (Han et al., 2018) and self-paced learning (Kumar et al., 2010).

We freeze the base weights of G , attach LoRA (Hu et al., 2022) to attention and MLP projections, and optimize **the weighted negative log-likelihood of F 's responses**

$$\mathcal{L}_{\text{FT}}(\Theta_{\text{LoRA}}) = \frac{1}{|\mathcal{H}|} \sum_{(S_t, a_t^F) \in \mathcal{H}} \omega_t \text{NLL}(a_t^F | G(S_t; \Theta_{\text{LoRA}})), \quad \omega_t \propto r_t. \quad (4)$$

The soft prompts are then discarded, and serving uses only the LoRA-adapted surrogate.

Before switching, SOMA verifies two conditions: the adapted surrogate must reproduce the original model's responses on a small acceptance batch, and the current query must remain close to the warm-start dialogue centroid. Once accepted, G serves later turns from a compressed input $\tilde{S}_t$ that keeps a fixed-budget summary of earlier content together with the most recent K turns, so the full history is never re-processed. During surrogate serving, SOMA keeps monitoring semantic drift. If recent queries move away from the learned local state, it rolls back to F , refreshes the summary and the centroid, and returns to G only after the acceptance gate passes again. **Because the drift statistic is a continuous distance to the warm-start centroid, gradual drift raises it progressively and triggers rollback once the margin is exhausted, rather than only on abrupt shifts.**

4 THEORETICAL ANALYSIS

This section answers two design questions: when SOMA may accept the session-local surrogate, and when its warm-start overhead is amortized. **Both rely on later turns staying tied to the warm-start**

state, not on their being easier, and characterize operating conditions rather than an advantage over other mining strategies. Proofs and a bound on the number of prompt candidates are in Appendix E.

4.1 SWITCHING CRITERION

At turn t , with shared context $S_t = \mathcal{D}_{t-1} \oplus q_t$, we measure the discrepancy of the adapted surrogate with the bounded score $\text{Gap}_\Theta(S_t) \in [0, 1]$ of Section 2, where Θ is the current adapter. **Consecutive turns are dependent**, so we assume that per-turn statistics form an m -dependent sequence and use the effective sample sizes $W_{\text{eff}} = W/(m+1)$ for the warm-start window and $|B|_{\text{eff}} = |B|/(m+1)$ for an acceptance batch B . Lemma E.1 shows that, with probability at least $1 - \delta$, a bounded warm-start statistic lies within $\eta_W(\delta) = \sqrt{\log(2/\delta)/(2W_{\text{eff}})}$ of its mean, so strongly dependent early turns call for a longer warm start before the switch can be assessed.

The fidelity gate evaluates the adapted surrogate on an acceptance batch B through its average gap $\hat{G}_B = \frac{1}{|B|} \sum_{S_j \in B} \text{Gap}_\Theta(S_j)$, and the surrogate passes when $\hat{G}_B$ is below a threshold ε .

Proposition 4.1 (Acceptance Bound for Switching). *Assume that warm-start and later states share the distribution $\mathcal{Q}_k$ (the locality premise of Section 2), let B be drawn from it and be disjoint from the turns used to fit Θ , and let $\gamma_B(\delta) = \sqrt{\log(1/\delta)/(2|B|_{\text{eff}})}$. With probability at least $1 - \delta$, $\mathbb{E}_{S \sim \mathcal{Q}_k}[\text{Gap}_\Theta(S)] \leq \hat{G}_B + \gamma_B(\delta)$. Hence $\hat{G}_B \leq \varepsilon - \gamma_B(\delta)$ certifies $\mathbb{E}[\text{Gap}_\Theta] \leq \varepsilon$.*

A union bound combines the two results (Corollary E.2), and an abrupt topic change fails the locality gate and triggers an immediate rollback to F . The guarantee has two practical limits. It needs a held-out acceptance batch, whereas our implementation reuses the warm-start turns for adaptation and acceptance, so in the reported experiments the gate is a heuristic check rather than a certified test. It is also loose: with $\delta = 0.05$ and $|B| = 24$, $\gamma_B \approx 0.25$, larger than the thresholds $\varepsilon \in [0.05, 0.12]$ themselves, which are therefore set empirically (Appendix E).

4.2 NET EFFICIENCY GAIN

Because SOMA pays its warm-start overhead once, its efficiency must be judged per session. Let T be the number of turns, W the switch point, and $c_F(\cdot)$ and $c_G(\cdot)$ the per-turn costs (latency, tokens, or price) of F and of G on its compressed input $\tilde{S}_t$; the net gain over serving every turn with F is

$$\Delta C(T; W) = \sum_{t=W+1}^T \left[c_F(\mathcal{D}_{t-1} \oplus q_t) - c_G(\tilde{S}_t) \right] - C_{\text{probe}}(W) - C_{\text{LoRA}}(W) - C_{\text{rb}}(T; W), \quad (5)$$

where C_{probe} and C_{LoRA} are the mining and adaptation costs and C_{rb} covers drift checks and rollback; SOMA pays off only when the accumulated post-switch savings exceed this one-time overhead. **This explains the operating range measured in Figure 3(a): sessions of 1–4 turns do not benefit, whereas locally coherent sessions of 5 or more turns amortize the initialization cost.**

5 EMPIRICAL STUDIES OF THE EFFECTIVENESS OF SOMA

We evaluate SOMA through five questions. **RQ1:** Can a small surrogate preserve the original model’s responses? **RQ2:** Does higher response similarity also improve task-grounded quality? **RQ3:** When does SOMA provide net efficiency gains after warm-start and adaptation overhead? **RQ4:** Which components of SOMA are most important? **RQ5:** When is session-local serving reliable, and when should SOMA fall back to the large model?

Datasets. We evaluate SOMA on six multi-turn benchmarks: ShareGPT (Chen et al., 2024a), ReMeDi (Yan et al., 2022), Craigslist (He et al., 2018), Multi-Char (agentlans, 2024), MATH (Hendrycks et al., 2021), and MT-Bench (Zheng et al., 2023). Dataset details and the context-dependence filtering are in Appendix C.1.

Models and baselines. We test two model families: LLaMA, using LLaMA-3.1-70B as F and LLaMA-2-7B as G , and Qwen, using Qwen-3-8B as F and Qwen-3-0.6B as G . We compare with Original, Surrogate, History-Prefix, History-FT, LLMLingua-2 (Pan et al., 2024), and RouteLLM (Ong et al., 2024); Appendix C.3 details each baseline.

Table 1: Response similarity to the original model on the LLaMA family. Higher is better. **SOMA is the closest to the original model on all six datasets, raising the average from 75.1 for the unadapted surrogate to 93.1 and exceeding the strongest baseline, RouteLLM, by 0.9 points.**

	ShareGPT	ReMeDi	Craigslist	Multi-Char	MATH	MT-Bench	Avg.
Surrogate	79.2 $\pm$ 2.18	82.7 $\pm$ 1.95	74.3 $\pm$ 2.36	70.8 $\pm$ 1.73	66.2 $\pm$ 2.91	77.5 $\pm$ 2.04	75.1 $\pm$ 5.98
History-Prefix	86.1 $\pm$ 1.67	87.9 $\pm$ 1.55	82.4 $\pm$ 1.72	84.7 $\pm$ 1.69	80.3 $\pm$ 3.83	87.2 $\pm$ 1.58	84.8 $\pm$ 2.94
History-FT	93.4 $\pm$ 2.12	91.8 $\pm$ 2.09	90.3 $\pm$ 1.23	89.1 $\pm$ 2.23	87.6 $\pm$ 2.26	92.4 $\pm$ 1.14	90.8 $\pm$ 2.18
LLMLingua-2	84.6 $\pm$ 1.72	86.4 $\pm$ 1.63	80.8 $\pm$ 1.91	82.9 $\pm$ 1.58	78.1 $\pm$ 2.77	85.3 $\pm$ 1.66	83.0 $\pm$ 3.11
RouteLLM	95.3 $\pm$ 1.44	92.5 $\pm$ 1.07	91.0 $\pm$ 1.86	91.4 $\pm$ 1.23	89.6 $\pm$ 1.95	93.2 $\pm$ 1.12	92.2 $\pm$ 1.98
SOMA	96.4 $\pm$ 1.91	93.2 $\pm$ 0.98	91.9 $\pm$ 2.49	92.3 $\pm$ 1.05	90.7 $\pm$ 1.12	94.1 $\pm$ 0.91	93.1 $\pm$ 1.99

Table 2: Response similarity to the original model on the Qwen family. Higher is better. **SOMA again leads on all six datasets; similarities are lower than for LLaMA because the size gap is wider, and SOMA’s average gain over the unadapted surrogate grows to 27.1 points.**

	ShareGPT	ReMeDi	Craigslist	Multi-Char	MATH	MT-Bench	Avg.
Surrogate	50.9 $\pm$ 1.21	63.5 $\pm$ 2.87	48.2 $\pm$ 3.40	44.7 $\pm$ 3.54	42.6 $\pm$ 1.70	40.4 $\pm$ 0.81	48.4 $\pm$ 8.31
History-Prefix	70.5 $\pm$ 1.15	75.7 $\pm$ 2.01	63.0 $\pm$ 2.23	65.5 $\pm$ 0.91	51.3 $\pm$ 2.45	53.6 $\pm$ 2.32	63.3 $\pm$ 9.47
History-FT	76.4 $\pm$ 1.68	79.2 $\pm$ 2.61	74.5 $\pm$ 3.56	63.2 $\pm$ 1.74	65.9 $\pm$ 1.70	62.7 $\pm$ 0.72	70.3 $\pm$ 7.23
LLMLingua-2	68.9 $\pm$ 1.32	74.1 $\pm$ 2.14	61.2 $\pm$ 2.67	63.4 $\pm$ 1.18	49.8 $\pm$ 2.03	51.7 $\pm$ 1.45	61.5 $\pm$ 9.49
RouteLLM	79.6 $\pm$ 1.04	81.9 $\pm$ 1.36	75.1 $\pm$ 2.91	72.8 $\pm$ 1.62	67.3 $\pm$ 1.21	68.0 $\pm$ 0.96	74.1 $\pm$ 5.95
SOMA	81.0 $\pm$ 0.93	83.2 $\pm$ 1.21	76.4 $\pm$ 3.85	74.2 $\pm$ 2.53	68.7 $\pm$ 1.28	69.2 $\pm$ 1.08	75.5 $\pm$ 5.97

Table 3: MATH exact-match accuracy. **SOMA stays below the original model but recovers 77–78% of the gap between the surrogate and the original model and outperforms every other baseline.**

Family	Orig.	Surr.	Hist.-P	Hist.-FT	Route	SOMA
LLaMA	48.34 $\pm$ 0.32	19.20 $\pm$ 0.78	25.03 $\pm$ 0.94	31.46 $\pm$ 0.87	33.88 $\pm$ 0.79	41.62 $\pm$ 0.66
Qwen	36.48 $\pm$ 0.41	11.73 $\pm$ 0.64	16.21 $\pm$ 0.79	22.57 $\pm$ 0.83	25.08 $\pm$ 0.71	31.14 $\pm$ 0.74

Metrics. We mainly measure teacher-fidelity similarity, i.e., how closely each method preserves the original model’s response, using three LLM judges: GPT-OSS (OpenAI, 2025), DeepSeek-V3, and Gemma-2-27B. To complement similarity with task-grounded quality, we report EM accuracy on MATH. Efficiency is measured by token usage, throughput, and end-to-end wall-clock latency, which for SOMA includes soft-prompt mining, LoRA adaptation, and gating. All models are served with vLLM 0.10.2 with automatic prefix caching enabled, so every baseline already benefits from KV-cache reuse across the turns of a session.

Implementation. We set the warm-start window, acceptance batch, candidate count, and switch threshold from Section 4; after switching, SOMA performs no further LoRA updates and keeps only the drift check. Appendix C.2 lists all hyperparameters.

5.1 EXPERIMENTAL RESULTS

RQ1: Response similarity to the original model. After the warm-start stage, the surrogate should preserve the original model’s behavior within the current dialogue state. Table 1 reports response similarity on the LLaMA family. SOMA achieves the highest average similarity across all six datasets. The gap over Surrogate shows the behavior loss caused by directly replacing the large model with a smaller model. The gap over History-FT shows that local adaptation helps, and becomes stronger when the warm-start turns are weighted by their anchoring scores. The largest gains appear on MATH and Multi-Char, where later turns depend on accumulated reasoning states, constraints, and roles. Table 2 repeats the evaluation on the Qwen family. Absolute similarities are lower because the capacity gap between Qwen-3-8B and Qwen-3-0.6B is wider, which makes local imitation harder, yet SOMA is again the best method on every dataset.

RQ2: Task-grounded quality. Response similarity is the primary metric for replacing the original model, but similarity alone does not guarantee task quality. We therefore evaluate MATH using exact-match accuracy. Table 3 shows that SOMA remains below the original large model, as expected, but substantially narrows the gap between the surrogate and the original model. SOMA also outperforms

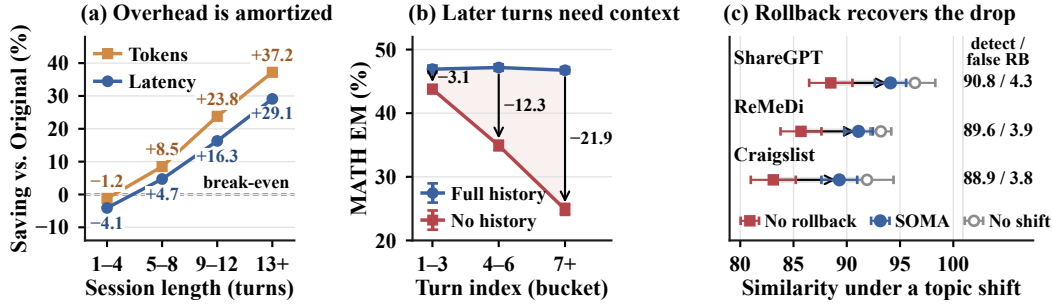


Figure 3: Operating range and reliability. (a) End-to-end saving of SOMA over Original on ShareGPT after all warm-start and adaptation overhead, by session length. (b) MATH exact match with and without the dialogue history: later turns remain context-dependent. (c) Response similarity under an abrupt topic shift without rollback, with SOMA’s drift-aware rollback, and without a shift; the right column lists SOMA’s shift-detection and false-rollback (RB) rates in %.

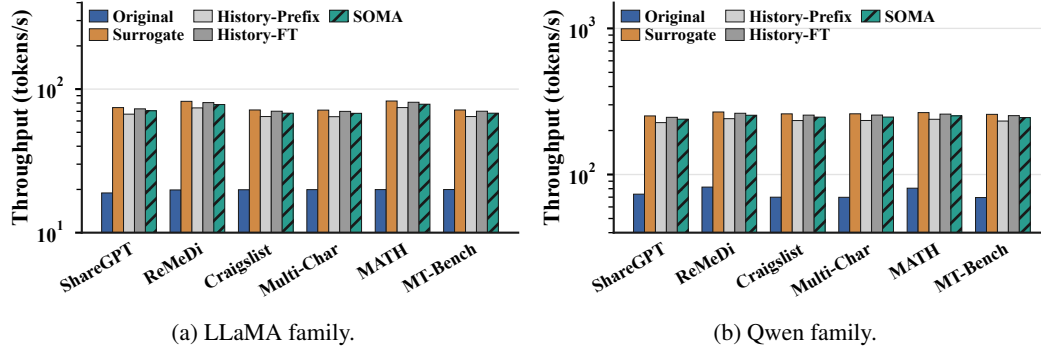


Figure 4: Throughput across six datasets. SOMA keeps the serving path close to the surrogate while maintaining higher similarity to the original model.

History-Prefix, History-FT, and RouteLLM in both model families. Thus SOMA’s gains are not only surface-level matching: local adaptation also preserves more of the original model’s reasoning.

RQ3: End-to-end efficiency. SOMA reduces post-switch cost by serving the adapted surrogate with compressed context, but it also introduces one-time costs from soft-prompt probing and LoRA adaptation. Thus, the key question is whether the total session cost decreases after this overhead is included. SOMA reduces average token usage by avoiding repeated full-history serving after switching (Appendix F.1). Figure 3(a) further shows the amortized operating range: **SOMA is slightly unfavorable for sessions of 1–4 turns (−4.1% latency), breaks even at 5–8 turns (+4.7%), and is clearly beneficial from 9 turns on (+16.3% to +29.1% latency, +23.8% to +37.2% tokens).** This matches SOMA’s intended use case as an amortized serving strategy for conversations where enough later turns remain local to the warm-start state. **Figure 4 reports throughput on all six datasets for both families: SOMA runs at 95–97% of the throughput of Surrogate and History-FT, 5.6% faster than History-Prefix, and 3.1–3.9× faster than Original (per-dataset token costs are in Appendix F.1).**

RQ4: Component contributions. We first ablate SOMA on the LLaMA family to identify which components drive the gains. Figure 5a shows that full SOMA performs best across all datasets. Removing the anti-degeneration loss consistently reduces performance, showing that entropy regularization keeps prompt mining stable and prevents collapsed, low-diversity soft prompts. Removing both the expectation-weighted term and the anti-degeneration loss leads to a larger drop, especially on harder datasets. **This result shows that the anchoring scores are informative only when the probe is both stable and semantic.** Figure 5b repeats the ablation on the Qwen family with the same ordering, so the conclusion does not depend on the backbone. Figure 5c breaks MT-Bench-101 down by ability for the LLaMA family: SOMA’s largest gains over the surrogate are on reasoning, reflection, and questioning, and it also improves understanding, interference handling, memory, and rephrasing, so its gains are not confined to a single skill.

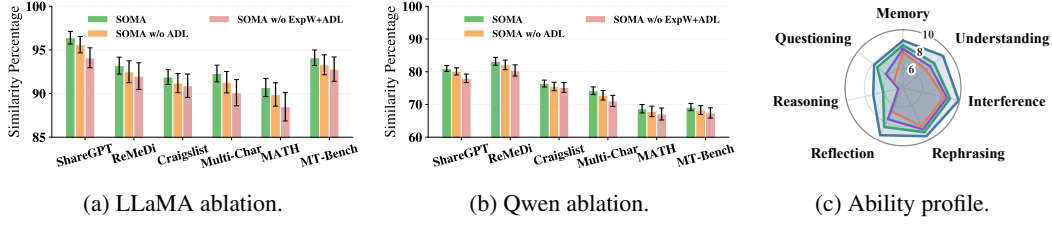


Figure 5: Component and capability analysis. Anti-degeneration regularization and expectation-weighted divergence both improve mining on LLaMA (a) and Qwen (b). (c) MT-Bench-101 ability profile of **Surrogate**, **History-Prefix**, **History-FT**, and **SOMA**.

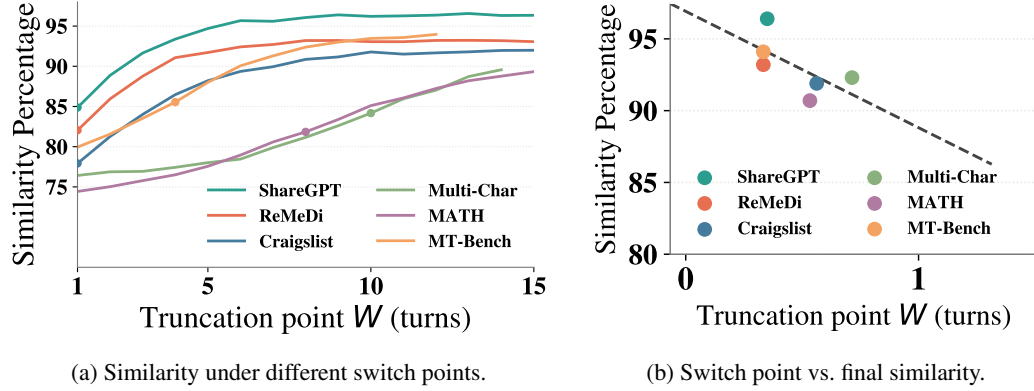


Figure 6: Warm-start and switching behavior. Easier local-dialogue settings switch earlier, while reasoning-heavy or multi-party settings need longer warm starts and achieve lower final similarity.

RQ5: Reliability and fallback. We next test when SOMA’s locality assumption holds and how rollback helps when it fails. Figure 3(b) shows that full-history MATH performance stays stable across turn buckets, while no-history performance drops sharply in later turns: **removing the history costs 3.1 EM points on turns 1–3 but 21.9 points on turns 7 and later**. This means later turns are not simply easier; they remain strongly dependent on the earlier dialogue context. Figure 3(c) then evaluates abrupt topic shifts. Without rollback, similarity drops substantially. With SOMA’s drift-aware rollback, **similarity returns to within 2.1–2.6 points of the clean setting (e.g., 88.5 → 94.1 vs. 96.4 on ShareGPT)**, with 88.9–90.8% of shifts detected and only 3.8–4.3% false rollbacks. These results show that SOMA should be viewed as a local serving strategy with automatic fallback. **Figure 6 shows the warm-start side: goal-anchored dialogues (ShareGPT, ReMeDi, Craigslist) reach high similarity after a few turns, whereas reasoning-heavy or multi-party ones (MATH, Multi-Char) need longer warm starts and plateau lower, so the switch point must be data-dependent, which is exactly what the acceptance gate implements.**

6 CONCLUSION

We presented SOMA, an efficient multi-turn LLM serving framework that uses early dialogue turns to adapt a small surrogate to the original model’s session-local behavior. **It probes the surrogate with adversarial soft prompts, distills the robustly anchored warm-start turns with LoRA, and switches only when alignment is sufficient.** Experiments show that SOMA better preserves original-model responses, improves task-grounded quality, and reduces serving cost once warm-start overhead is amortized. These results suggest that session-local surrogate serving is a practical direction for efficient multi-turn LLM deployment **when sessions are long and locally coherent enough to amortize the warm-start cost, as Figure 3(a) quantifies.** Because SOMA only consumes the original model’s text outputs, it also applies when F is a proprietary model behind an API, provided that the surrogate’s embedding layer is accessible for mining. In practice, SOMA is most useful for sessions that run beyond a handful of turns on a single task, and it degrades gracefully otherwise, because the gate keeps short or drifting sessions on the original model. We discuss the limitations of SOMA and directions for future work in Appendix I.

AI USE STATEMENT

In this work, we used generative AI tools to improve the readability of the manuscript, such as grammar checking, wording refinement, and formatting suggestions, and to assist with implementing and debugging parts of the experimental code, such as evaluation and plotting scripts. We did not use generative AI tools to propose or refine the research hypotheses, design the methodology or the experiments, develop the theoretical framework, formulate or prove the mathematical claims, generate or clean datasets, or interpret the results. We have reviewed all AI-assisted work: AI-suggested edits were checked by the authors for accuracy and faithfulness to the intended meaning, and AI-assisted code was verified and tested for correctness by the authors before use. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies efficient multi-turn LLM serving and does not involve human subjects, user studies, or the collection of new personal data. All experiments use publicly available dialogue and reasoning benchmarks under their respective licenses (Appendix C.1), and we release no new dataset. SOMA is intended to reduce the cost and latency of serving LLMs, which can broaden access to interactive LLM systems and reduce their energy footprint. As with any method that substitutes a smaller surrogate for a larger model, there is a risk of quality degradation when a conversation drifts away from the adapted local region, which is most consequential in high-stakes domains such as medical, legal, or financial assistance. SOMA mitigates this risk through its semantic acceptance gate and drift-aware rollback, but real deployments should retain monitoring, conservative switching thresholds, and human oversight where errors may cause harm. Because SOMA adapts to dialogue history, deployments should also follow data-minimization and retention practices to protect user privacy. Broader impacts and limitations are discussed further in Appendices H and I.

REPRODUCIBILITY STATEMENT

We have made efforts to ensure that our results are reproducible. An anonymized implementation of SOMA, including scripts for all major experiments, is available at <https://anonymous.4open.science/r/SOMA-4175>. Appendix C.1 describes the datasets and Appendix C.4 the filtering of non-context-dependent dialogues; Appendix C.2 gives the implementation details and hyperparameter ranges of SOMA; Appendix C.3 specifies the baselines; Appendix C.5 describes the LLM-judge protocol; and Appendix C.6 lists the software packages and versions used. For the theoretical results in Section 4, all assumptions are stated explicitly and complete proofs are provided in Appendix E, with the supporting mathematical background in Appendix D.

REFERENCES

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- agentlans. Multi-character dialogue dataset. Hugging Face Dataset, 2024. URL <https://huggingface.co/datasets/agentlans/multi-character-dialogue>. CC-BY-4.0 license.
- Shalini Ananda. Beyond the bubble: How context-aware memory systems are changing the game in 2025. Tribe AI Applied AI Blog, April 2025.
- Anthropic. Introducing claude. Anthropic Blog, March 2023. <https://www.anthropic.com/news/introducing-claude>.
- Anthropic. Prompt caching with claude. <https://www.anthropic.com/news/prompt-caching>, August 2024. Accessed 2025-06-12.
- Ge Bai, Jie Liu, Xingyuan Bu, Yancheng He, Jiaheng Liu, Zhanhui Zhou, Zhuoran Lin, Wenbo Su, Tiezheng Ge, Bo Zheng, et al. Mt-bench-101: A fine-grained benchmark for evaluating large language models in multi-turn dialogues. *arXiv preprint arXiv:2402.14762*, 2024.

- Adarsh Prasad Behera, Jaya Prakash Champati, Roberto Morabito, Sasu Tarkoma, and James Gross. Towards efficient multi-llm inference: Characterization and analysis of llm routing and hierarchical techniques. *arXiv preprint arXiv:2506.06579*, 2025.
- Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural computation*, 15(6):1373–1396, 2003.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Conghui He, Jiaqi Wang, Feng Zhao, and Dahua Lin. Sharegpt4v: Improving large multi-modal models with better captions. In *European Conference on Computer Vision*, pp. 370–387. Springer, 2024a.
- Nuo Chen, Hongguang Li, Juhua Huang, Baoyuan Wang, and Jia Li. Compress to impress: Unleashing the potential of compressive memory in real-world long-term conversations. *arXiv preprint arXiv:2402.11975*, 2024b.
- Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Ruhle, Laks VS Lakshmanan, and Ahmed Hassan Awadallah. Hybrid llm: Cost-efficient and quality-aware query routing. *arXiv preprint arXiv:2404.14618*, 2024.
- Xin Luna Dong, Seungwhan Moon, Yifan Ethan Xu, Kshitiz Malik, and Zhou Yu. Towards next-generation intelligent assistants leveraging llm techniques. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 5792–5793, 2023.
- Bin Gao, Zhuomin He, Puru Sharma, Qingxuan Kang, Djordje Jevdjic, Junbo Deng, Xingkun Yang, Zhou Yu, and Pengfei Zuo. {Cost-Efficient} large language model serving for multi-turn conversations with {CachedAttention}. In *2024 USENIX Annual Technical Conference (USENIX ATC 24)*, pp. 111–126, 2024.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. Hipporag: Neurobiologically inspired long-term memory for large language models. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- Bo Han, Quanming Yao, Xingrui Yu, Gang Niu, Miao Xu, Weihua Hu, Ivor Tsang, and Masashi Sugiyama. Co-teaching: Robust training of deep neural networks with extremely noisy labels. In *Advances in Neural Information Processing Systems*, 2018.
- He He, Derek Chen, Anusha Balakrishnan, and Percy Liang. Decoupling strategy and generation in negotiation dialogues. *arXiv preprint arXiv:1808.09637*, 2018.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. *arXiv preprint arXiv:1904.09751*, 2019.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- Svante Janson. Large deviations for sums of partly dependent random variables. *Random Structures & Algorithms*, 24(3):234–248, 2004.

- Jinwoo Jeong and Jeongseob Ahn. Accelerating llm serving for multi-turn dialogues with efficient resource management. In *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2*, pp. 1–15, 2025.
- M. Pawan Kumar, Benjamin Packer, and Daphne Koller. Self-paced learning for latent variable models. In *Advances in Neural Information Processing Systems*, 2010.
- Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. Llms get lost in multi-turn conversation. *arXiv preprint arXiv:2505.06120*, 2025.
- Haoyang Li, Zhanchao Xu, Yiming Li, Xuejia Chen, Darian Li, Anxin Tian, Qingfa Xiao, Cheng Deng, Jun Wang, Qing Li, Lei Chen, and Mingxuan Yuan. Loopserve: An adaptive dual-phase llm inference acceleration system for multi-turn dialogues. *arXiv preprint arXiv:2507.13681*, 2025a.
- Huayang Li, Tian Lan, Zihao Fu, Deng Cai, Lemao Liu, Nigel Collier, Taro Watanabe, and Yixuan Su. Repetition in repetition out: Towards understanding neural text degeneration from the data perspective. *Advances in Neural Information Processing Systems*, 36:72888–72903, 2023.
- Lincan Li, Zheng Chen, and Yushun Dong. Llm as clinical graph structure refiner: Enhancing representation learning in eeg seizure diagnosis, 2026. URL <https://arxiv.org/abs/2604.28178>.
- Yubo Li, Xiaobin Shen, Xinyu Yao, Xueying Ding, Yidi Miao, Ramayya Krishnan, and Rema Padman. Beyond single-turn: A survey on multi-turn interactions with large language models. *arXiv preprint arXiv:2504.04717*, 2025b.
- Fang Liu, Yang Liu, Lin Shi, Houkun Huang, Ruifeng Wang, Zhen Yang, Li Zhang, Zhongqi Li, and Yuchi Ma. Exploring and evaluating hallucinations in llm-powered code generation. *arXiv preprint arXiv:2404.00971*, 2024.
- Xiang Liu, Hong Chen, Xuming Hu, and Xiaowen Chu. Flowkv: Enhancing multi-turn conversational coherence in llms via isolated key-value cache management. *arXiv preprint arXiv:2505.15347*, 2025.
- Clara Meister, Tiago Pimentel, Gian Wiher, and Ryan Cotterell. Locally typical sampling. *Transactions of the Association for Computational Linguistics*, 11:102–121, 2023.
- Eric Melz. Enhancing llm intelligence with arm-rag: Auxiliary rationale memory for retrieval augmented generation. *arXiv preprint arXiv:2311.04177*, 2023.
- Don Moon. Accelerating multi-turn llm serving with multi-tier caching and smarter scheduling. Medium (Byte-Sized AI), April 2025.
- Dilxat Muhtar, Yelong Shen, Yaming Yang, Xiaodong Liu, Yadong Lu, Jianfeng Liu, Yuefeng Zhan, Hao Sun, Weiwei Deng, Feng Sun, Xueliang Zhang, Jianfeng Gao, Weizhu Chen, and Qi Zhang. Streamadapter: Efficient test time adaptation from contextual streams. *arXiv preprint arXiv:2411.09289*, 2024.
- Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M Waleed Kadous, and Ion Stoica. Routellm: Learning to route llms with preference data, 2024. URL <https://arxiv.org/abs/2406.18665>.
- OpenAI. gpt-oss-120b & gpt-oss-20b model card, 2025. URL <https://arxiv.org/abs/2508.10925>.
- Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor Rühle, Yuqing Yang, Chin-Yew Lin, et al. Llmilingua-2: Data distillation for efficient and faithful task-agnostic prompt compression. *arXiv preprint arXiv:2403.12968*, 2024.
- Soya Park and Chinmay Kulkarni. Thinking assistants: Llm-based conversational assistants that help users think by asking rather than answering. *arXiv preprint arXiv:2312.06024*, 2023.
- Sam T Roweis and Lawrence K Saul. Nonlinear dimensionality reduction by locally linear embedding. *science*, 290(5500):2323–2326, 2000.

- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36:68539–68551, 2023.
- Jay Shah, Ganesh Bikshandi, Ying Zhang, Vijay Thakkar, Pradeep Ramani, and Tri Dao. Flashattention-3: Fast and accurate attention with asynchrony and low-precision. *Advances in Neural Information Processing Systems*, 37:68658–68685, 2024.
- Ying Sheng, Shiyi Cao, Dacheng Li, Coleman Hooper, Nicholas Lee, Shuo Yang, Christopher Chou, Banghua Zhu, Lianmin Zheng, Kurt Keutzer, Joseph E. Gonzalez, and Ion Stoica. S-lora: Serving thousands of concurrent lora adapters. *arXiv preprint arXiv:2311.03285*, 2023.
- Tal Shnitzer, Anthony Ou, Mírian Silva, Kate Soule, Yuekai Sun, Justin Solomon, Neil Thompson, and Mikhail Yurochkin. Large language model routing with benchmark datasets. *arXiv preprint arXiv:2309.15789*, 2023.
- Andreas Stolcke, Klaus Ries, Noah Coccaro, Elizabeth Shriberg, Rebecca Bates, Daniel Jurafsky, Paul Taylor, Rachel Martin, Carol Van Ess-Dykema, and Marie Meteer. Dialogue act modeling for automatic tagging and recognition of conversational speech. *Computational linguistics*, 26(3): 339–373, 2000.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Qingyue Wang, Yanhe Fu, Yanan Cao, Shuai Wang, Zhiliang Tian, and Liang Ding. Recursively summarizing enables long-term dialogue memory in large language models. *Neurocomputing*, 639: 130193, 2025.
- Yan Wang, Dongyang Ma, and Deng Cai. With greater text comes greater necessity: Inference-time training helps long text generation. *arXiv preprint arXiv:2401.11504*, 2024.
- Sean Welleck, Ilya Kulikov, Stephen Roller, Emily Dinan, Kyunghyun Cho, and Jason Weston. Neural text generation with unlikelihood training. *arXiv preprint arXiv:1908.04319*, 2019.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks, 2024. URL <https://arxiv.org/abs/2309.17453>, pp. 1, 2024.
- Guojun Yan, Jiahuan Pei, Pengjie Ren, Zhaochun Ren, Xin Xin, Huasheng Liang, Maarten De Rijke, and Zhumin Chen. Remedi: Resources for multi-domain, multi-service, medical dialogues. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 3013–3024, 2022.
- Zihao Yi, Jiarui Ouyang, Yuwen Liu, Tianhao Liao, Zhe Xu, and Ying Shen. A survey on recent advances in llm-based multi-turn dialogue systems. *arXiv preprint arXiv:2402.18013*, 2024.
- Dahai Yu, Lin Jiang, Rongchao Xu, and Guang Wang. Healthmamba: An uncertainty-aware spatiotemporal graph state space model for effective and reliable healthcare facility visit prediction, 2026.
- Haozhen Zhang, Tao Feng, and Jiaxuan You. Router-r1: Teaching llms multi-round routing and aggregation via reinforcement learning. *arXiv preprint arXiv:2506.09033*, 2025.
- Tianhao Zhang, Jie Yang, Deli Zhao, and Xinliang Ge. Linear local tangent space alignment and application to face recognition. *Neurocomputing*, 70(7-9):1547–1553, 2007.
- Yuying Zhao, Yu Wang, Xueqi Cheng, Anne Marie Tumlin, Yunchao Liu, Damin Xia, Meng Jiang, and Tyler Derr. Amplifying your social media presence: Personalized influential content generation with llms. *arXiv preprint arXiv:2505.01698*, 2025.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.

APPENDIX

A NOTATIONS

This section summarizes all notations used throughout this paper.

Table 4: Notation summary.

Symbol	Meaning
$\mathcal{D}_k$	Dialogue prefix $\{(q_t, a_t)\}_{t=1}^k$
$q_{\leq t}$	Full history up to turn t
$\bar{F}, G$	Original (large) model; surrogate (small) model
S_t	Dialogue state $\mathcal{D}_{t-1} \oplus q_t$ at turn t
$\mathcal{Q}_k$	Distribution of later dialogue states after $\mathcal{D}_k$ (local region)
$\mathcal{V}, E$	G 's vocabulary; embedding matrix $E \in \mathbb{R}^{d \times \mathcal{V} }$
Π_t	Next-token distribution of G at turn t ; entry $\Pi_t[v]$
$\mathbf{e}_v$	Embedding of token v (column of E)
$\mathbf{P} \in \mathbb{R}^{L \times d}$	Soft prompt (row-wise prompt tokens in embedding space)
$N_k(u)$	k nearest neighbors of token u by cosine in E
$\bar{\mathbf{e}}_{t,i}$	Expected embedding $\bar{\mathbf{e}}_{t,i} = E^\top \Pi_{t,i}$ at position i
$w_{t,i}$	Expectation-weighted factor for semantic divergence loss
$\text{Gap}_G(S)$	Response discrepancy $1 - \cos(\phi(a^G(S)), \phi(a^F(S))) \in [0, 1]$
r_t	Anchoring score of warm-start turn t (residual mining loss)
$ B _{\text{eff}}$	Effective batch size $ B /(m+1)$ under m -dependence
$\hat{G}_B$	Average gap of the adapted surrogate on a held-out acceptance batch

B RELATED WORK

Efficient LLM for Multi-turn Dialogues. Recent efforts to improve multi-turn LLM serving efficiency fall into two main paradigms. The first involves **single-model approaches** that reduce context length or reuse computation. These include *summarization and context compression* (Wang et al., 2025; Chen et al., 2024b; Xiao et al., 2024), *memory augmentation* (Melz, 2023; Gutiérrez et al., 2024), and *caching and attention reuse*, such as **CachedAttention** (Gao et al., 2024), **prompt caching** (Anthropic, 2024), **resource-aware multi-turn scheduling** (Jeong & Ahn, 2025), **isolated per-turn KV compression in FlowKV** (Liu et al., 2025), and **adaptive prefill sparsification with progressive KV compression in LoopServe** (Li et al., 2025a). While effective in lowering per-turn cost, these methods still rely on repeated large-model inference, which also leads to high API expenses, latency, and GPU demand. Moreover, they may truncate or underutilize dialogue context, harming performance on complex tasks. The second paradigm adopts **multi-model approach**, using smaller models for simple queries and escalating difficult ones to larger LLMs (Behera et al., 2025; Schick et al., 2023; Ding et al., 2024), typically via model routing (Shnitzer et al., 2023; Ong et al., 2024), **multi-round routing and aggregation learned with reinforcement learning** (Zhang et al., 2025), and distillation (Hinton et al., 2015). However, pre-trained small models often generalize poorly on complex multi-turn dialogues, while switching models adds inefficiency.

Relation to systems-level serving and test-time adaptation. Prefix and KV caching (CachedAttention, prompt caching, FlowKV, LoopServe) reduce the prefill cost of re-processing the history on the *same* large model; the per-token decode cost, memory footprint, and API price of that model are unchanged. SOMA is orthogonal: after the switch it removes the large model from the serving path, and the surrogate can itself be served with prefix caching. Our efficiency comparison already serves every method, including Original, through vLLM with automatic prefix caching enabled (Section 5), so the reported token and latency gains are measured on top of KV-cache reuse. Test-time adaptation methods such as Temp-LoRA (Wang et al., 2024) and StreamAdapter (Muhtar et al., 2024) adapt a model to its *own* generated or in-context text; History-FT and SOMA instead perform cross-model distillation, adapting the surrogate to the large model's warm-start responses within the session.

Local Manifold Approximation. We use “local manifold approximation” in the informal sense of classical local methods such as LLE (Roweis & Saul, 2000), LTSA (Zhang et al., 2007), and Laplacian eigenmaps (Belkin & Niyogi, 2003): a dialogue prefix induces a local region of dialogue states, and SOMA approximates the large model only within that region rather than globally. SOMA does not compute intrinsic dimensions, tangent spaces, or geodesics, and it does not match the hidden spaces of the two models. The local view enters the method only through the output-space discrepancy of Problem 2.1, which compares responses through a shared sentence encoder, and through the locality gate. Its measurable prediction, that warm-start supervision transfers to later turns until the topic shifts, is tested empirically in Section 5 and Appendix F.2.

C REPRODUCIBILITY

In this section, we introduce the details of the experiments in this paper for reproducibility. At the same time, we have uploaded all necessary code to our GitHub repository to reproduce the results presented in this paper: <https://anonymous.4open.science/r/SOMA-4175>. All major experiments are encapsulated as shell scripts, which can be conveniently executed. We introduce the details for reproducibility in the subsections below.

C.1 REAL-WORLD DATASETS

In this section, we briefly present the real-world datasets used in this paper, and all these datasets are commonly used datasets in multi-turn conversation tasks. *ShareGPT* (Chen et al., 2024a) is a large-scale collection of high-quality image–text conversations and captions. *ReMeDi* (Yan et al., 2022) is a multi-domain Chinese medical dialogue corpus of doctor–patient conversations. *Craigslist* (He et al., 2018) contains multi-turn buyer–seller negotiation chats from Craigslist, enabling study of bargaining strategies and goal-directed dialogue. *Multi-Char* (agentlans, 2024) provides multi-character conversational scenarios with role specifications to evaluate coordination and role consistency in multi-party dialogue. *MATH* (Hendrycks et al., 2021) is a benchmark of competition-style math problems with step-by-step solutions designed to assess mathematical reasoning in language models. *MT-Bench* (Zheng et al., 2023) is a multi-turn benchmark to assess response quality across diverse tasks. In this study, we filter out the non-context-dependent dialogues in these datasets.

C.2 IMPLEMENTATION OF SOMA

We implement SOMA based on PyTorch with HuggingFace Transformers, serve inference via vLLM with FlashAttention (Shah et al., 2024), and run on **Nvidia A6000 GPUs**. Soft-prompt tuning optimizes only the prompt tensor $\mathbf{P} \in \mathbb{R}^{L \times d}$ on the surrogate G using AdamW, cosine decay, gradient clipping, and a KV cache for a single forward pass per turn. The objective combines unlikelihood on a semantic neighborhood, an expectation-weighted penalty using the truncated top- m expectation, and a light anti-degeneration entropy term. **The warm-start context–response pairs, weighted by their anchoring scores r_t , are then used to adapt G with LoRA on attention and MLP projections (rank r), keeping the base weights frozen; early stopping is triggered by validation divergence. At inference, a cosine closeness gate with a lightweight sentence encoder decides a one-time switch to the adapted surrogate. All hyperparameters below, including W , $|B|$, M , and (k, m) , are set empirically within the listed ranges; the bounds in Section 4 describe how the required margins scale but are too loose at these sample sizes to set them directly.** Prompt length $L \in \{4, 8, 16, 32, 64\}$; learning rate $\eta \in [1 \times 10^{-4}, 5 \times 10^{-3}]$; AdamW weight decay $\lambda_{\text{wd}} \in [0, 10^{-2}]$; gradient clip $\in [0.5, 1.0]$; neighborhood size $k \in \{20, 50, 100\}$; expectation truncation $m \in \{50, 100, 200\}$; temperature $\tau \in [0.4, 1.2]$; expectation weight $\lambda \in [0.5, 2.0]$; anti-degeneration weight $\beta \in [0.02, 0.15]$; LoRA rank $r \in \{4, 8, 16, 32\}$ with scale $\alpha_{\text{lor}} \in \{4, 8, 16, 32\}$; LoRA LR $\eta_{\text{lor}} \in [5 \times 10^{-5}, 2 \times 10^{-3}]$; warm-start window $W \in [3, 12]$ turns; acceptance batch $|B| \in [6, 24]$ contexts; parallel candidates $M \in \{3, 4, 5\}$; switch threshold $\varepsilon \in [0.05, 0.12]$ with $m_{\text{cons}} \in \{2, 3\}$.

C.3 IMPLEMENTATION OF BASELINES

Original: We query the large model (family-appropriate: LLaMA-3.1-70B or Qwen-3-8B) with the full dialogue history at every turn to get the responses.

You are an expert conversation analyst.

Task – Classification:

* **context-dependent***: The response of every turn relies on information from earlier turns (the conversation is a single chain of related reasoning steps).

* **context-independent***: The response of every turn is independent of earlier turns (the conversation is a collection of independent questions and answers).

Conversation (verbatim):

{dialogue}

Question: Is the conversation context-dependent or context-independent?

Please answer with exactly one word:

Either **context-dependent** or **context-independent** — nothing else.

Figure 7: Instructions for the LLM to filter context-dependent dialogue.

Surrogate: We query the small model (LLaMA-2-7B or Qwen-3-0.6B) with the full dialogue history at every turn to get the responses.

History-Prefix: The surrogate receives the entire conversation produced by the Original up to $t-1$ and generates a_t ; no parameter updates are performed. The switching is the same as SOMA.

History-FT: We fine-tune the surrogate on (S_t, a_t^F) pairs where S_t is the Original’s full context up to t and a_t^F is the next reply, using LoRA on attention projections with early stopping; inference then runs the fine-tuned surrogate without the Original. The switching is the same as SOMA.

LLMLingua-2 (Pan et al., 2024): We compress the Original’s history at each turn with LLMLingua-2 and feed the compressed summary to the surrogate; we follow the authors’ recommended settings.

RouteLLM (Ong et al., 2024): We adopt the released router to choose between the Original and the surrogate per turn (complex vs. simple queries), following the original settings from the paper.

C.4 DATA FILTERING

To ensure that SOMA is evaluated on context-dependent dialogues, we prompt multiple strong LLMs, including GPT-OSS, DeepSeek-V3, and Gemma-2-27B, with the classification prompt shown in Figure 7, and retain only conversations that all three models recognize as context-dependent. This avoids model-specific biases and yields a high-quality subset of dialogues where later turns meaningfully depend on earlier turns, matching the use cases in this paper.

C.5 IMPLEMENTATION OF LLM JUDGE

The prompt for the LLM judge is shown in Figure 8. It adapts the single-answer grading protocol of MT-Bench (Zheng et al., 2023) and MT-Bench-101 (Bai et al., 2024) to reference-based similarity scoring: the judge receives the original model’s response as the reference and scores the surrogate’s response on a 0–1 scale along accuracy, reasoning, context integration, and communication. The reference-based framing and the 0–1 scale are our modifications; the MT-Bench protocols grade single answers on a 1–10 scale without a reference. Each response is scored by three judges (GPT-OSS, DeepSeek-V3, and Gemma-2-27B), and we report the mean.

You are an impartial evaluator.
 You are evaluating the AI assistant’s capabilities in solving a problem posed by ‘Human’.
 Assess the assistant’s performance based on the following criteria:

1. Accuracy: Verify the correctness of the AI assistant’s answer against the ground truth (reference solution).
2. Reasoning: Assess the completeness, clarity, and logical soundness of the step-by-step reasoning process.
3. Context Integration: Consider whether the assistant integrates relevant prior dialogue or context that influences the solution.
4. Communication: Appraise how clearly and effectively the assistant communicates its reasoning to aid understanding.

Score Guidelines (0.0 to 1.0):

- 1.0: Completely correct answer and meticulously clear, step-by-step reasoning that is logically sound and instructional.
- 0.7–0.9: Correct answer with a well-articulated reasoning process that includes the necessary steps and promotes understanding.
- 0.4–0.6: Partially correct answer with some minor reasoning flaws or omissions.
- 0.0–0.3: Incorrect answer and/or poor reasoning that lacks clarity or logic.

Only respond with a single score between 0.0 and 1.0. Do not include any explanation.

Reference Solution:
 {Original Response}
 Candidate Answer:
 {Surrogate Response}

Score:

Figure 8: Instructions for the LLM judge to evaluate the response similarity.

C.6 PACKAGES REQUIRED FOR IMPLEMENTATION

We perform all experiments on a server equipped with Nvidia A6000 GPUs. Below we list the key packages and their versions used in our implementation:

- **Python** == 3.10
- **pytorch** == 2.8.0 + CUDA 12.8
- **torchvision** == 0.19.0
- **torchaudio** == 2.8.0
- **numpy** == 1.26.x
- **pandas** == 2.2.x
- **scipy** == 1.12.x
- **cmake** == 3.28+
- **ninja** == 1.11+
- **ipython** == 8.x
- **psutil** == 5.9+
- **vllm** == 0.10.2
- **transformers** == 4.56.1
- **accelerate** == 1.9.0
- **bitsandbytes** == 0.46.1
- **sentencepiece** == 0.2.0
- **tiktoken** == 0.11.0
- **einops** == 0.8.1

- **datasets** == 4.0.0
- **huggingface-hub** == 0.34.2
- **safetensors** == 0.5.3
- **ray** == 2.49.1
- **scikit-learn** == 1.7.1
- **fastapi** == 0.116.2
- **uvicorn** == 0.35.0

D MATHEMATICAL CONCEPTS

Here we provide a more comprehensive view of the relevant concepts that can be helpful to understand the idea of stratification as discussed in the main body of the paper.

Definition D.1 (Preimage). Let $f : X \mapsto Y$ be a function from a set X (*domain*) to a set Y (*codomain*). For any subset $N \subseteq Y$, the *preimage* of N under f , denoted $f^{-1}(N)$, is defined as:

$$f^{-1}(N) = \{x \in X \mid f(x) \in N\}.$$

In other words, $f^{-1}(N)$ consists of all elements in the domain X that are mapped into the subset N of the codomain Y , regardless of whether every element of N has a preimage.

Definition D.2 (Metric Space). A set X , whose elements are called points, is said to be a *metric space* if for any two points $p, q \in X$, there is an associated real number $d(p, q)$, called the *distance* from p to q , such that the following three conditions hold:

1. $d(p, q) \geq 0$, and $d(p, q) = 0 \iff p = q$;
2. $d(p, q) = d(q, p)$ (symmetry);
3. $d(p, q) \leq d(p, r) + d(r, q)$ for any $r \in X$ (triangle inequality).

Any function satisfying these properties is called a *distance function*, or a *metric*.

Definition D.3 (Neighborhood). Let X be a metric space. A set $N_r(p) \subset X$ is called a *neighborhood* of a point $p \in X$ if it consists of all points $q \in X$ such that $d(p, q) < r$ for some radius $r > 0$. The number r is called the *radius* of the neighborhood.

Definition D.4 (Continuous). Let X and Y be two topological spaces. A function $f : X \mapsto Y$ is *continuous* if for each point $x \in X$ and each neighborhood N of $f(x)$ in Y , the set $f^{-1}(N)$ is a neighborhood of the point $x \in X$ in the domain.

Definition D.5 (Topological Equivalence or Homeomorphism). A function $h : X \mapsto Y$ is called a *homeomorphism* if it is one-to-one, continuous, and has a continuous inverse function. When such a function exists, X and Y are called *homeomorphic* (or *topologically equivalent*) spaces.

Definition D.6 (Open Set). A subset $U \subseteq X$ is called an *open set* if for every point $p \in U$, there exists a neighborhood $N_r(p) \subseteq U$. That is, each point in U has some "wiggle room" around it that still lies entirely within the set U itself.

Definition D.7 (Countable Base (Second Countability)). Let X be a topological space. A collection $\mathcal{B}$ of open subsets of X is called a *base* (or *basis*) for the topology on X if for every open set $U \subseteq X$ and every point $x \in U$, there exists a set $B \in \mathcal{B}$ such that

$$x \in B \subseteq U.$$

If there exists a base $\mathcal{B}$ that is *countable*, then the space X is said to be *second countable* or to have a *countable base*; Euclidean spaces, with balls of rational radii around rational points, are a standard example.

Definition D.8 (Hausdorff Space). A topological space with the property that two distinct points can always be surrounded by disjoint open sets is called a *Hausdorff space*.

Essentially, Hausdorff spaces are the spaces where any two points being "far off" is defined.

Definition D.9 (Manifold). A *manifold* of dimension n is a second-countable Hausdorff topological space in which each point has a neighborhood homeomorphic to Euclidean space $\mathbb{R}^n$.

Definition D.10 (Smooth Manifold). A *smooth manifold* is a manifold $\mathcal{M}$ equipped with a collection of coordinate charts (i.e., homeomorphisms $\varphi : U \rightarrow \mathbb{R}^n$ for open sets $U \subseteq \mathcal{M}$) such that all *transition maps* between overlapping charts,

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j),$$

are infinitely differentiable (i.e., C^∞). This structure is known as a *smooth atlas*, and it allows calculus to be performed on the manifold.

E THEORETICAL RESULTS AND PROOFS

This appendix states and proves the results used in Section 4. Each statement lists its assumptions, and Section 4.1 discusses where our implementation departs from them.

Dependence and effective sample size. Consecutive dialogue turns are not independent. We assume that the per-turn statistics $Z_1, \dots, Z_N$ form an m -dependent sequence, that is, $\{Z_s\}_{s \leq t}$ and $\{Z_s\}_{s > t+m}$ are independent for every t , and we define the effective sample size $N_{\text{eff}} = N/(m+1)$; independent samples correspond to $m = 0$. For such sequences Hoeffding’s inequality holds with N replaced by N_{eff} : if each Z_s takes values in an interval of length R , then for every $u > 0$

$$\Pr \left(\frac{1}{N} \sum_{s=1}^N (Z_s - \mathbb{E}Z_s) \geq u \right) \leq \exp \left(- \frac{2N_{\text{eff}} u^2}{R^2} \right),$$

and the same bound holds for the lower tail. This follows from Janson’s inequality for sums of partly dependent variables (Janson, 2004), applied to the dependency graph of the sequence, whose fractional chromatic number is at most $m+1$.

Locality premise. The results below compare statistics measured on warm-start turns with their expectations over the later turns of the session. We therefore assume, as in Section 2, that warm-start states and later states share the marginal distribution $\mathcal{Q}_k$. This is the premise that the locality gate monitors and that Figures 3(c) and 6 test; otherwise they describe only the warm-start distribution.

Switching Criterion: Warm-Start Concentration and Acceptance

Lemma E.1 (Warm-Start Concentration). Let $h(S) \in [0, 1]$ be a local statistic that is fixed before the window is observed, such as $\text{Gap}_\Theta(S)$ for a fixed adapter Θ or the distance of a query to a fixed centroid. Let $h(S_1), \dots, h(S_W)$ be m -dependent with common mean $h^* = \mathbb{E}_{S \sim \mathcal{Q}_k}[h(S)]$, and write $\hat{h}_W = \frac{1}{W} \sum_{t=1}^W h(S_t)$. Then, with probability at least $1 - \delta$,

$$|\hat{h}_W - h^*| \leq \eta_W(\delta) := \sqrt{\frac{\log(2/\delta)}{2W_{\text{eff}}}}.$$

Proof of Lemma E.1. Apply the m -dependent Hoeffding inequality with $R = 1$ to each tail at level $\delta/2$. Setting $\exp(-2W_{\text{eff}}u^2) = \delta/2$ gives $u = \eta_W(\delta)$, and a union bound over the two tails gives the claim. $\square$

Statement: Proposition 4.1. Let the adapter Θ be fixed given the adaptation data, and let the acceptance batch $B = \{S_1, \dots, S_{|B|}\}$ be disjoint from that data and drawn from $\mathcal{Q}_k$ as an m -dependent sequence. Let $\hat{G}_B = \frac{1}{|B|} \sum_j \text{Gap}_\Theta(S_j)$ and $G^* = \mathbb{E}_{S \sim \mathcal{Q}_k}[\text{Gap}_\Theta(S)]$. Then, with probability at least $1 - \delta$,

$$G^* \leq \hat{G}_B + \gamma_B(\delta), \quad \gamma_B(\delta) = \sqrt{\frac{\log(1/\delta)}{2|B|_{\text{eff}}}}.$$

In particular, if $\hat{G}_B \leq \varepsilon - \gamma_B(\delta)$, then $G^* \leq \varepsilon$ with probability at least $1 - \delta$.

Proof of Proposition 4.1. Condition on the adaptation data. Because B is disjoint from it, Θ is fixed, and $\text{Gap}_\Theta(S_1), \dots, \text{Gap}_\Theta(S_{|B|})$ are m -dependent with mean G^* and range $R = 1$. The lower-tail inequality gives $\Pr(\widehat{G}_B - G^* \leq -u) \leq \exp(-2|B|_{\text{eff}}u^2)$, and setting the right-hand side to δ gives $u = \gamma_B(\delta)$. On the complementary event, $G^* < \widehat{G}_B + \gamma_B(\delta) \leq \varepsilon$ if $\widehat{G}_B \leq \varepsilon - \gamma_B(\delta)$. $\square$

When the acceptance batch is reused. If B overlaps the adaptation data, Θ depends on the S_j in B . The gaps then no longer have mean G^* given Θ , $\widehat{G}_B$ is biased downward by the fit, and Proposition 4.1 does not apply. Our implementation reuses the warm-start turns in this way (Appendix G); holding out part of the warm-start window would restore the bound at the cost of fewer adaptation examples. The bound is also loose at the sample sizes used here: for $\delta = 0.05$, $m = 0$, and $|B| = 24$, $\gamma_B = 0.25$, which already exceeds the thresholds $\varepsilon \in [0.05, 0.12]$. It therefore describes how the required margin scales with $|B|_{\text{eff}}$ rather than setting the thresholds.

Corollary E.2 (Switching Rule). *Let the locality statistic h satisfy the conditions of Lemma E.1, and let B satisfy those of Proposition 4.1. If $\widehat{G}_B \leq \varepsilon - \gamma_B(\delta)$ and $\hat{h}_W \leq \tau - \eta_W(\delta)$, then, with probability at least $1 - 2\delta$, both $G^* \leq \varepsilon$ and $h^* \leq \tau$.*

Proof of Corollary E.2. By Lemma E.1, $h^* \leq \hat{h}_W + \eta_W(\delta) \leq \tau$ with probability at least $1 - \delta$. By Proposition 4.1, $G^* \leq \varepsilon$ with probability at least $1 - \delta$. A union bound shows that both hold with probability at least $1 - 2\delta$. $\square$

Coverage and Suboptimality to Determine the Number of Soft-Prompt Candidates

Assume that the mining objective is locally quadratic with curvature $\mathbf{H}_T \succeq 0$ whose top eigenvector $\mathbf{v}_1$ lies in an r_{act} -dimensional active subspace. SOMA draws its M candidate prompts from an isotropic Gaussian, so their projections onto this subspace point in independent, uniformly random directions. The results below bound the probability that one of them starts within angle θ of $\mathbf{v}_1$, and the fraction of the top curvature that such a direction retains. They concern the initial directions rather than the optimized prompts, and we set $M \in \{3, 4, 5\}$ empirically (Appendix C.2).

Proposition E.3 (Coverage of the Best Local Direction). *Let $r_{\text{act}} \geq 2$, let $\mathbf{v}_1$ be a unit vector in an r_{act} -dimensional subspace, and draw $\mathbf{u}_1, \dots, \mathbf{u}_M$ i.i.d. from $\text{Unif}(\mathbb{S}^{r_{\text{act}}-1})$ in that subspace. For $\theta \in (0, \pi/2]$, let*

$$p_{r_{\text{act}}}(\theta) = \Pr(\angle(\mathbf{u}_1, \mathbf{v}_1) \leq \theta) = \frac{1}{2} I_{\sin^2 \theta} \left(\frac{r_{\text{act}}-1}{2}, \frac{1}{2} \right),$$

where $I_x(\cdot, \cdot)$ is the regularized incomplete beta function. Then

$$\Pr \left(\min_{m \leq M} \angle(\mathbf{u}_m, \mathbf{v}_1) \leq \theta \right) = 1 - (1 - p_{r_{\text{act}}}(\theta))^M, \quad p_{r_{\text{act}}}(\theta) \geq \frac{(\sin \theta)^{r_{\text{act}}-1}}{(r_{\text{act}}-1) B(\frac{r_{\text{act}}-1}{2}, \frac{1}{2})}.$$

For example, $p_{r_{\text{act}}}(\pi/2) = 1/2$ for every r_{act} . Isotropic Gaussian initializations satisfy the sampling assumption, because the normalized projection of $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ onto a fixed subspace is uniformly distributed on the unit sphere of that subspace.

Proof of Proposition E.3. Step 1: Exact cap probability. Write $r = r_{\text{act}}$ and $T = \langle \mathbf{u}_1, \mathbf{v}_1 \rangle$. For $\mathbf{u}_1$ uniform on $\mathbb{S}^{r-1}$, T is symmetric about 0 and $T^2 \sim \text{Beta}(\frac{1}{2}, \frac{r-1}{2})$, so $1 - T^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{1}{2})$. For $\theta \leq \pi/2$ we have $\cos \theta \geq 0$, hence

$$\Pr(T \geq \cos \theta) = \frac{1}{2} \Pr(T^2 \geq \cos^2 \theta) = \frac{1}{2} \Pr(1 - T^2 \leq \sin^2 \theta) = \frac{1}{2} I_{\sin^2 \theta} \left(\frac{r-1}{2}, \frac{1}{2} \right).$$

Step 2: Lower bound. Let $a = (r-1)/2$ and $x = \sin^2 \theta$. Since $(1-s)^{-1/2} \geq 1$ on $[0, 1)$,

$$I_x(a, \frac{1}{2}) = \frac{1}{B(a, \frac{1}{2})} \int_0^x s^{a-1} (1-s)^{-1/2} ds \geq \frac{1}{B(a, \frac{1}{2})} \int_0^x s^{a-1} ds = \frac{x^a}{a B(a, \frac{1}{2})},$$

so $p_r(\theta) \geq x^a / (2a B(a, \frac{1}{2})) = (\sin \theta)^{r-1} / ((r-1) B(\frac{r-1}{2}, \frac{1}{2}))$.

Step 3: Independence across M samples. The M draws are independent, so none of them falls in the cap with probability $(1 - p_r(\theta))^M$, which gives the coverage probability. $\square$

Lemma E.4 (Directional Suboptimality). *Let $\mathbf{H}_T \succeq 0$ have largest eigenvalue λ_1 and top eigenvector $\mathbf{v}_1$. For any unit $\hat{\mathbf{u}}$ with $\angle(\hat{\mathbf{u}}, \mathbf{v}_1) \leq \theta \leq \pi/2$,*

$$\hat{\mathbf{u}}^\top \mathbf{H}_T \hat{\mathbf{u}} \geq \lambda_1 \cos^2 \theta, \quad \lambda_1 - \hat{\mathbf{u}}^\top \mathbf{H}_T \hat{\mathbf{u}} \leq \lambda_1 \sin^2 \theta.$$

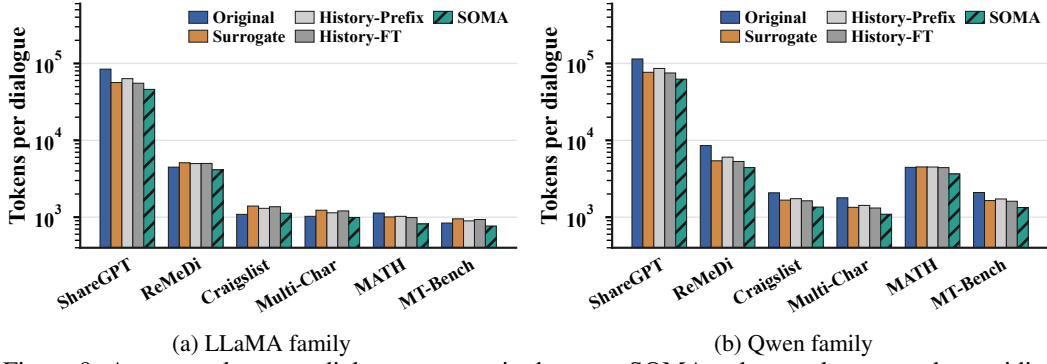


Figure 9: Average tokens per dialogue across six datasets. SOMA reduces token usage by avoiding repeated full-history serving after switching.

Proof of Lemma E.4. Let $\varphi = \angle(\hat{\mathbf{u}}, \mathbf{v}_1) \leq \theta$ and decompose $\hat{\mathbf{u}} = \cos \varphi \mathbf{v}_1 + \sin \varphi \mathbf{w}$ with $\|\mathbf{w}\|_2 = 1$ and $\mathbf{w} \perp \mathbf{v}_1$. Because $\mathbf{H}_T \mathbf{v}_1 = \lambda_1 \mathbf{v}_1$, the cross term vanishes, and

$$\hat{\mathbf{u}}^\top \mathbf{H}_T \hat{\mathbf{u}} = \lambda_1 \cos^2 \varphi + \sin^2 \varphi \sum_{j \geq 2} \lambda_j \langle \mathbf{v}_j, \mathbf{w} \rangle^2 \geq \lambda_1 \cos^2 \varphi \geq \lambda_1 \cos^2 \theta,$$

since $\lambda_j \geq 0$ and $\cos^2$ is decreasing on $[0, \pi/2]$. Subtracting from λ_1 gives the residual bound. $\square$

Corollary E.5 (Candidate Budget). *Under the conditions of Proposition E.3, the coverage probability within angle θ is at least $1 - \delta$ whenever $M \geq \log(1/\delta) / \log(1/(1 - p_{\text{rct}}(\theta)))$. Because $\log(1/(1 - p)) \geq p$, the simpler condition $M \geq (r_{\text{act}} - 1) B(\frac{r_{\text{act}} - 1}{2}, \frac{1}{2}) \log(1/\delta) / (\sin \theta)^{r_{\text{act}} - 1}$ also suffices. Together with Lemma E.4, this trades candidate count against the quality of the best direction.*

Proof of Corollary E.5. Coverage fails with probability $(1 - p)^M$, where $p = p_{\text{rct}}(\theta)$, and $(1 - p)^M \leq \delta$ holds if and only if $M \log(1/(1 - p)) \geq \log(1/\delta)$. The second condition follows from $\log(1/(1 - p)) \geq p$ and the lower bound on p in Proposition E.3. $\square$

F ADDITIONAL EXPERIMENTAL RESULTS

This appendix provides supporting results that complement the main experiments: dataset-level token costs, the effect of warm-start length on switching, and the empirical variance concentration behind the switching bound in Section 4.1.

F.1 ADDITIONAL EFFICIENCY RESULTS

The main text reports the amortized break-even analysis and throughput. Here we provide the dataset-level token results for both model families. Figure 9 shows that **SOMA uses the fewest tokens per dialogue on 11 of the 12 dataset-family pairs; the exception is Craigslist with LLaMA, where Original uses 3% fewer**. The reason is that after switching, SOMA serves later turns with the adapted surrogate and a compressed context rather than repeatedly sending the full growing history. This avoids the large token cost of Original and History-Prefix while retaining stronger response similarity than directly using the unadapted surrogate.

F.2 WARM-START LENGTH AND SWITCHING BEHAVIOR

The main text (Figure 6) reports how final similarity depends on the switch point. The sweep uses warm-start windows $W \in [1, 15]$ and measures the final response similarity after switching. Once the curve flattens, adding more warm-start turns brings little quality gain but delays the efficiency benefit, so short warm starts suffice for general or goal-anchored chats, while compositional reasoning and multi-party dialogues need more evidence before switching.

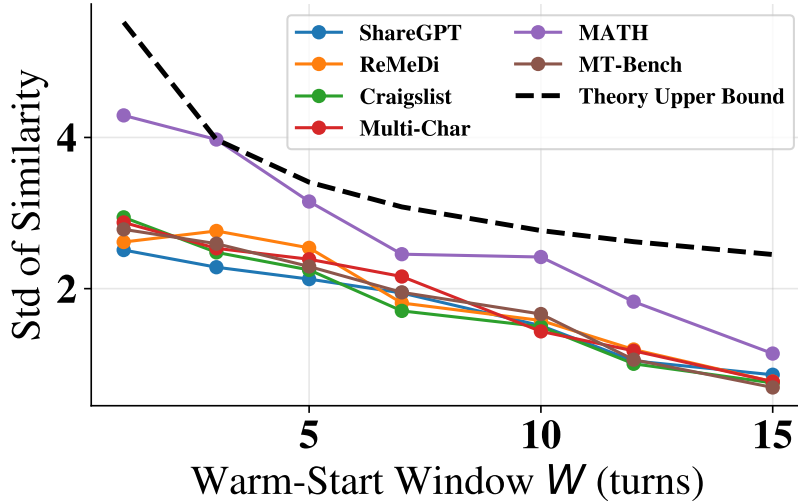


Figure 10: Variance of response similarity versus warm-start window W . Larger warm-start windows reduce variance and support more stable switching decisions.

F.3 EMPIRICAL VARIANCE CONCENTRATION

Finally, we examine how the variance of surrogate–teacher similarity changes with the warm-start window W . For each dataset, we compute the empirical standard deviation across dialogues at each truncation point. Figure 10 shows that variance decreases as W grows: early turns are highly variable, while larger warm-start windows enter a more stable regime. This trend is consistent with the concentration behavior used in Lemma E.1 and the switching bound in Proposition 4.1. In other words, early turns provide the largest information gain, while later warm-start turns mainly reduce uncertainty until the switching decision becomes stable.

G DESIGN RATIONALE AND SERVING CONSIDERATIONS

This section collects design questions raised in review that the main text only touches on.

Choice of soft prompts and of an adversarial objective. Soft prompts are a convenient probe for three practical reasons: they are differentiable, so a handful of gradient steps on the surrogate G can search a continuous neighborhood of the warm-start context instead of enumerating discrete paraphrases; they live in G ’s own embedding space, so the semantic neighborhood $\mathcal{N}_k(\cdot)$ and the expected-embedding weight are computed with the same token embeddings and reuse the logits of a single forward pass; and the ANN-based neighborhood keeps the per-step cost proportional to $k + m$ rather than $|\mathcal{V}|$. Alternative perturbations (input noise, paraphrasing, or adversarial token substitution) can also explore beyond the finite warm-start turns, but they either require an external paraphraser or a discrete search, and they do not directly expose which *directions* in G ’s input space make G diverge from F . We push G away from F rather than toward it because the score must measure how robust the agreement is, not how much agreement a prompt can add. A constructive (divergence-minimizing) prompt would raise the agreement on every turn and yield a signal confounded with the capacity of the prompt itself; an adversarial prompt removes the agreement that is easy to remove, and r_t records what survives. The prompts are then discarded, and only the anchoring scores r_t enter the LoRA objective in Eq. 4 as per-turn training weights.

Relation to divergence-weighted History-FT and Random-FT. The LoRA objective in Eq. 4 weights every warm-start turn by its anchoring score $\omega_t \propto r_t$, so SOMA can be read as a weighted History-FT whose weights come from prompt probing rather than from the raw output divergence between G and F on the warm-start turn. The two weightings measure different things and point in opposite directions: raw output divergence emphasizes turns that the *unperturbed* surrogate currently gets wrong, whereas the anchoring score emphasizes turns whose teacher response the surrogate

still predicts *after* an adversarial perturbation, i.e., responses that are determined by the dialogue state and hence likely to transfer to later turns. The ablation in Figure 5a isolates the contributions of the mining objective’s two terms, but it does not include a baseline that weights History-FT by raw output divergence or one that samples training turns uniformly at random with the same LoRA budget; we consider both key controls and leave them, with a prompt-length sweep, to future work.

Robustness, divergence, and the perturbation space. We do not prove that robustness under prompt perturbations is a better predictor of adaptation utility than direct output divergence, nor that the continuous perturbation space of $\mathbf{P}$ covers the space of natural-language future turns. We therefore treat these as modeling assumptions and test them indirectly: the topic-shift stress test (Figure 3(c)) probes future turns that leave the mined region, and the warm-start sweep (Appendix F.2) shows how much of the local behavior is captured as the window grows. A direct empirical comparison of the two weightings is the most informative next step for validating the mining objective.

Number of large-model calls and target consistency. The original model F is queried once per warm-start turn to serve the user; these W responses are stored and reused as the mining targets a_t^F , as the LoRA targets, and as the references for the acceptance gate. F is not re-queried while $\mathbf{P}$ is optimized, so the targets are never stale with respect to the dialogue state, and the soft prompt perturbs only the surrogate side of the objective. Because the acceptance gate reuses these turns, the acceptance bound of Section 4.1 does not certify the gate as implemented; holding out part of the warm-start window would restore it at the cost of fewer adaptation examples. Prompt mining and LoRA adaptation run entirely on the surrogate and consist of a small number of forward–backward passes over the W warm-start turns (M candidate prompts times the mining iterations, followed by the LoRA steps). Their wall-clock cost is part of the end-to-end latency reported in Figure 3(a).

Drift detection. The drift statistic is the cosine distance between recent queries and the warm-start centroid, monitored over a sliding window; rollback triggers after m_{cons} consecutive violations. Because the statistic is continuous, gradual drift lowers it progressively rather than only at abrupt shifts, and Corollary E.2 would bound the probability of accepting a surrogate whose true gap exceeds the margin if the acceptance batch were held out and the centroid fixed in advance. As implemented, both come from the warm-start turns, so the gate is a heuristic check (Section 4.1), and it does not certify that the centroid distance tracks every form of quality degradation, which is why we also report the false-rollback and detection rates in Figure 3(c).

Serving many session-local adapters. A LoRA adapter of rank r on the attention and MLP projections of LLaMA-2-7B has about $2.5r$ million parameters ($\approx 20\text{M}$ parameters, or roughly 40 MB in bfloat16, for $r = 8$), i.e., well under 1% of the surrogate. Multi-adapter serving systems batch requests from many adapters over a single copy of the base weights and page adapters between host and GPU memory (Sheng et al., 2023), so GPU residency is bounded by the number of concurrently *switched* sessions rather than by the total number of sessions, and an adapter is discarded when its session ends or rolls back permanently. The mining and adaptation steps for a session are independent of other sessions and can be scheduled as short background jobs on the surrogate replica. We did not benchmark multi-node concurrent adapter scheduling; the reported latencies are for the single-session serving path described in Section 3.

H BROADER IMPACT

SOMA aims to reduce the cost and latency of multi-turn LLM serving by replacing repeated large-model inference with session-local surrogate serving. This can make interactive LLM systems more efficient, accessible, and environmentally sustainable. At the same time, using a smaller surrogate may introduce quality degradation if the dialogue leaves the learned local state, especially in high-stakes settings such as medical, legal, or financial assistance. SOMA mitigates this risk through semantic acceptance and drift-aware rollback, but deployment should still include monitoring, conservative thresholds, and human oversight where errors may cause harm. Since SOMA uses dialogue history for local adaptation, practical deployments should also apply data minimization, retention limits, and privacy-preserving processing of the dialogue history.

I LIMITATIONS AND FUTURE WORK

SOMA has several limitations. First, the surrogate’s capacity limits how closely it can match the original model. When the small–large model gap is very large, some behaviors cannot be recovered by session-local adaptation alone. Second, soft-prompt mining requires access to the surrogate tokenizer and embedding space, which makes SOMA harder to deploy in strict black-box API settings. Third, SOMA is designed for locally coherent dialogues. Abrupt topic changes, noisy warm-start windows, or strong paraphrase shifts can weaken the learned local directions, although the drift gate is designed to reduce persistent degradation. Finally, SOMA introduces a one-time warm-start cost, so it is most suitable for medium-to-long sessions where post-switch savings can amortize the initial adaptation overhead. Two further limitations concern evaluation. Gradual topic drift is detected by the same continuous centroid statistic as abrupt shifts, but we have only stress-tested abrupt shifts (Figure 3(c)); quality may degrade before the rollback margin is exhausted. We also have not compared SOMA against a History-FT variant weighted by raw output divergence or against uniformly sampled local turns under the same LoRA budget (Appendix G), so the incremental value of prompt mining over simpler weightings is supported by the component ablation rather than by a direct head-to-head comparison against these alternatives.

Future work can improve SOMA along several directions. Better drift detectors could make rollback more reliable under topic shifts. Approximate mining methods could reduce the need for internal surrogate access. Multi-region adaptation could support conversations with multiple topics, and privacy-preserving mining could make the method safer for sensitive dialogue settings. Another promising direction is to amortize soft-prompt search across related sessions and extend SOMA to multimodal serving, where early turns often carry images or documents.